



11372 - Is the NIR Counterpart of FRB 20250316A a Transient?

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRCam Imaging	(1) FRB20250316A

ABSTRACT

Fast radio bursts (FRBs) remain one of the most mysterious phenomena in time-domain astrophysics. While improved localization capabilities have confirmed their extragalactic origin, their diverse host galaxy environments have not shed direct light on their progenitor systems. Indeed, numerous models have emerged, involving compact objects like magnetars in various configurations, including with binary companions. Deep counterpart searches at other wavelengths provide the most promising path to uncovering FRB emission mechanisms and progenitor systems; however, such searches require precisely-localized and nearby FRBs. Recently, CHIME/FRB localized FRB 20250316A to a precision of 11×13 pc in the galaxy NGC4141 at $d = 40$ Mpc, making it a unique testbed for counterpart models. A subsequent JWST observation, obtained 2 months after the burst, revealed a faint NIR source at the location of the FRB, the first potential non-radio counterpart of an extragalactic FRB. Here we propose a repeated JWST observation at least a year later, when transient emission will have faded, to conclusively determine if the source is a transient or persistent counterpart. The detection of transient IR emission, likely due to a dust echo of a magnetar flare, would be a groundbreaking discovery with direct insight into the emission mechanism, rapidly advancing the FRB field forward.

OBSERVING DESCRIPTION

We will use NIRCам to obtain a second epoch of deep imaging of FRB 20250316A with the extra-wide F150W2 and F322W2 filters to test for fading of the NIR counterpart detected in the first epoch of JWST observations. We will use a 16-point sub-pixel dither pattern, 10 groups per exposure, FULL array of Module B, and the MEDIUM8 readout mode.

Proposal 11372 - Targets - Is the NIR Counterpart of FRB 20250316A a Transient?

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	FRB20250316A	RA: 12 09 44.3190 (182.4346625d) Dec: +58 50 56.71 (58.84909d) Equinox: J2000	Epoch of Position: 2000.0	
Comments: Category=Unidentified Description=[Infrared sources]					

Proposal 11372 - Observation 1 - Is the NIR Counterpart of FRB 20250316A a Transient?

Observation	Proposal 11372, Observation 1										Thu Apr 30 18:00:09 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCам Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	FRB20250316A	RA: 12 09 44.3190 (182.4346625d) Dec: +58 50 56.71 (58.84909d) Equinox: J2000			Epoch of Position: 2000.0					
	Comments: Category=Unidentified Description=[Infrared sources]										
Template	Module					Subarray					
	B					FULL					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRASCA		4		SMALL-GRID-DITHER		8" (SMALL)		3	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W2	F322W2	MEDIUM8	7	2	24	12	17651.25		
Special Requirements	Offset 38.8 arcsec, 38.3 arcsec										